

Things I Don't Understand About Quantum Error Correction

Tree-based vs surface codes — the questions that only make sense inside the frame

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Poster Session: Wed 26 Aug, 16:30–18:00 · Rowan Brad Quni-Gudzinas · QNFO

Panel 1 — The premise I reject

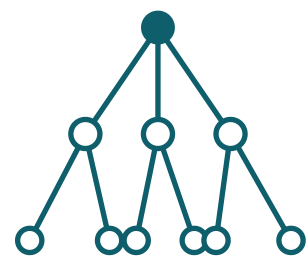
QEC is not necessary in well-designed hardware. Nature runs uncorrected, robust quantum processes — photosynthesis, magnetoreception — with no correction scaffold at all. The correction burden is a design choice, priced by the physical cost of the machine you built.

NISQ is an imperfect bridge to better physics. Even on the bridge, there are better QEC options than surface codes — tree-based codes are one. The hierarchy I am testing:

- **Punch-card era:** bug-fix; correction collapses as reliability improves.
- **Active QEC:** surface / LDPC — the correction loop on noisy hardware.
- **Tree-based class:** nested, distinction-based ultrametric trees — structural protection, no active loop (Bruhat–Tits T_2 one instance among many).
- **Topological:** TQC — protection designed into the topology, not applied.

The spine: "Don't mistake physics for algorithm." The fragility of the substrate is fixed; the code architecture is a free choice. TQC does not improve the physics — it changes the math. So does the tree class.

Panel 2 — The comparison: tree-based vs surface codes



One instance of the class: the Bruhat–Tits tree T_2 .
The class: nested, distinction-based ultrametric trees.

Surface codes (the standard): 2D lattice; active syndrome measurement; classical decoder; local stabilizers; MWPM over Euclidean distances; $d \sim \sqrt{N}$; proven threshold $\sim 1\%$; cheap parallel decode; $\sim 10^3\text{--}10^4$ physical qubits per logical.

Tree-based codes (the alternative): hierarchical, nonlocal, recursive; nested distinction-based structures; BP+OSD decode level-by-level; constant rate with d linear in N — the "good-code" regime the programme names (quantum Tanner codes); no proven threshold yet; needs high-degree / long-range connectivity.

All numbers are classical simulation. No tree code has run on hardware.

Error model (qubit, code-capacity)	Tree	Surface	Ratio
Bit-flip	50.0%	$\sim 10.9\%$	4.6×
Depolarizing	75.0%	$\sim 1.0\%$	75×
Independent $X+Z$	17.30%	$\sim 18.9\%$	0.92×

Depolarizing uses the code's structured-noise channel (not yet published); surface comparators are standard published figures. Under independent errors the qudit generalization sits at $\approx 2 \times 10^{-4}$ — about 55× below surface codes; the qubit claim targets correlated-failure regimes.

Bound (Tanner counting): rate $\geq 1 - 5b/5c$ — high-degree hierarchical graphs (the tree class) carry the rate that flat lattices cannot. The open question: does hierarchical decoding amortize better than the $\sqrt{N}$ blow-up of surface codes?

Panel 3 — Things I don't understand (1–4)

1 · What is the lowest achievable joules-per-correct-answer for a fault-tolerant machine on a fixed workload — and is it ever below the classical machine's?

JPCUB's core question. Not answerable by definition: the field has never priced it. Computed here (model + output in the repo): on a trapped-ion machine, the energy-optimal code is qLDPC $[[144,12,12]]$ with erasure conversion — $\sim 36\times$ less energy per correct answer than surface $d=11$ at $p_2=1e-3$ (~ 0.96 J) vs ~ 35 J per 10^4 correct gates). The tree class's regime is correlated failure, not this one.

2 · Does the erasure bill (kT·ln2 per correction) grow with code sophistication, or is there a code family whose per-answer cost drops as redundancy rises?

Every syndrome reset and ancilla re-init is a Landauer erasure. More redundancy = more erasure — unless structure replaces erasure. I cannot find the erasure bill priced anywhere.

3 · Is there a code whose correction overhead vanishes as the hierarchy deepens — does structural protection converge to nature's zero-correction regime, or is there a floor?

Photosynthesis and magnetoreception run robust quantum processes with no correction scaffold. The floor, if any, is a thermodynamic fact; I have not seen it measured.

4 · Are tree-structured / ultrametric codes a provably different complexity class of decoding (hierarchical vs metric-matching), or merely a relabeling of LDPC?

BP+OSD on a tree is recursive; MWPM on a surface is metric. "Provably different" would be a structural fact about the code class — not a relabeling.

Panel 4 — Things I don't understand (5–8)

5 · Which quantum behaviors are substrate-intrinsic (cannot be engineered away) vs algorithm-chosen (designable)? What is the limit of structure-provided protection?

TQC proves topology helps. The limit of structure-provided protection is the boundary between physics and algorithm — nobody draws it.

6 · Is there a quantum speedup that no classical algorithm can match, independent of hardware — or is "quantum advantage" hostage to a classical-computing model that has never been interrogated?

The bit is a lossy quantization. If the classical model is itself a choice, the comparison is a comparison of two choices, not of a machine against a law.

7 · What does it cost to measure the syndrome itself — the observer is inside the code?

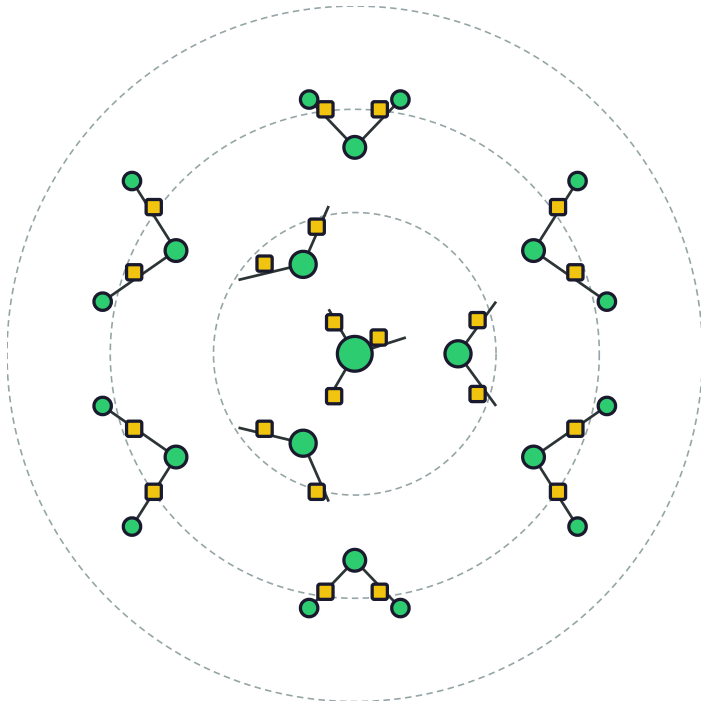
Nobody prices the measurement scaffold's own energy. "Reality is a syndrome; we probe it, we never read it directly" — the observer is a node in the code.

8 · Does "quantum" survive an intrinsic definition — not "not-classical", not "tensor of bits"? What is a quantum system in itself?

Every definition in the room was circular. The field's identity is definitionally unsettled — that is data, not a deficit.

The test for a real unknown: if the answer were either value, the field's behavior would have to change. A tautology's answer changes nothing.

Panel 5 — The ZX picture — the tree code in the field's diagrammatic language



Spiders — the atom.
Green = Z-basis copy/merge; red = X-basis. In the graph-state convention: green spider (0 in, 1 out) = $|+\rangle$; yellow H-box (2 in, 2 out) = CZ = the edge.

Pauli webs — the stabilizers, read off the diagram.
A connected sub-diagram that evaluates to a Pauli operator. The code's stabilizers and logical operators are paths through the spider web.

Gadgets — the field's cost unit.
Phase/Pauli gadgets $e^{\pm i\alpha P}$; T-count optimization is gadgetization — the algebraic cousin of an energy cost. This poster prices the thermodynamic one.

Codeword — the protected state; the map the code guards.

Syndrome — the symptom pattern; a probe that never reveals the state itself.

Tree-structured code — a nested-ball / ultrametric hierarchy with hierarchical decoding.

Panel 6 — What would change my mind

The erasure engine: "Error correction is a Landauer machine; every correction erases information and pays $kT\cdot\ln 2$. Nature's uncorrected quantum processes avoid the erasure bill entirely — structure instead of erasure. JPCUB prices the difference."

A matched-noise comparison: the tree family vs a surface code under the same correlated-noise model, in classical simulation. If the advantage falls to parity, the comparison on this poster is answered.

A decoder reality check: does a known decoder family already cover tree-structured codes, and how does its complexity scale?

An independent re-run: the simulator and the protocol are handed over to anyone who wants to run the comparison themselves.

The classical test, today: these codes already guard your flash RAM (NAND P/E wear is a physical cost). A tree-structured nested-ball code guarding flash better and cheaper is falsifiable now — no quantum hardware needed. JPCUB is general, not quantum-only.

Numeric criteria: matched-noise parity $\Rightarrow$ the advantage claim is withdrawn · a decoder that already covers trees $\Rightarrow$ ask 2 answered · an independent re-run that fails $\Rightarrow$ the simulation claim is withdrawn. Every criterion is stated with its number.

"We compute on discrete matrices yet insist reality is a continuous Hilbert space. Which is the real model?"